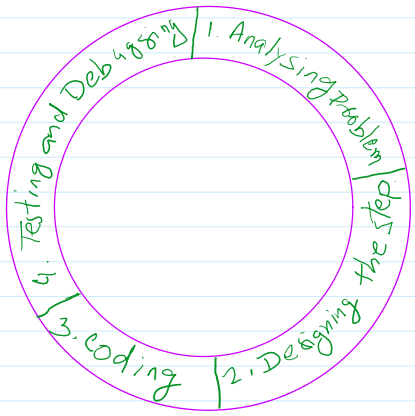
Introduction to Problem Solving

24 August 2024 17:22

Steps for problem solving.



Steps for problem solving

Algorithms : steps to solve a problem

For example:- You need to find out GCD or HCF of two number.

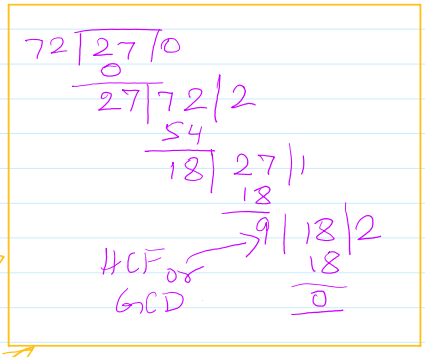
HCF (27, 72) = 9

27 = 3 × 3 × 3

72 = 2 × 2 × 2 × 3 × 3

3 × 3 = 9

Aryabhatta



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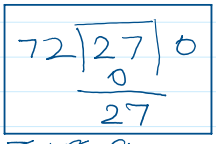


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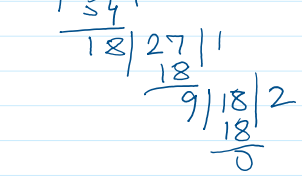
1.  INPUT "TWO NUMBER"; X, Y
2.  WHILE Y > 0
3.      R ← (X MOD Y)
4.      X ← Y
5.      Y ← R
6.  PRINT "HCF"; X
7.  END
```

X = ~~27~~ ~~72~~ ~~27~~ ~~18~~ 9

Y = ~~72~~ ~~27~~ ~~18~~ ~~9~~ 0

R = ~~27~~ ~~18~~ ~~9~~ 0



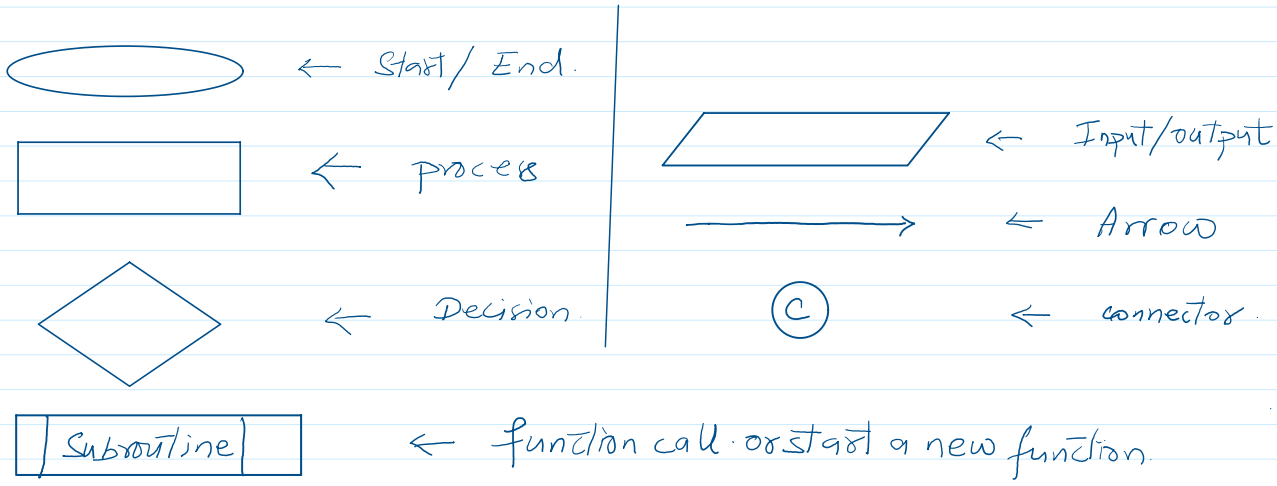
Why do we need Algorithm?

A programmer writes a program to instruct the computer to do certain tasks as desired. The computer then follows the steps written in the program code. Therefore, the programmer first prepares a road map of the program to be written before actually writing the code Without a road map, the programmer may not be able to clearly visualize the instructions to be written, and may end up developing a program which may not work as expected. Such a road map is nothing but the algorithm, which is the building block of a computer program. For example, searching using a search engine, sending a message, finding a word in a document, booking a taxi through an app, performing online banking, playing computer games, are based on

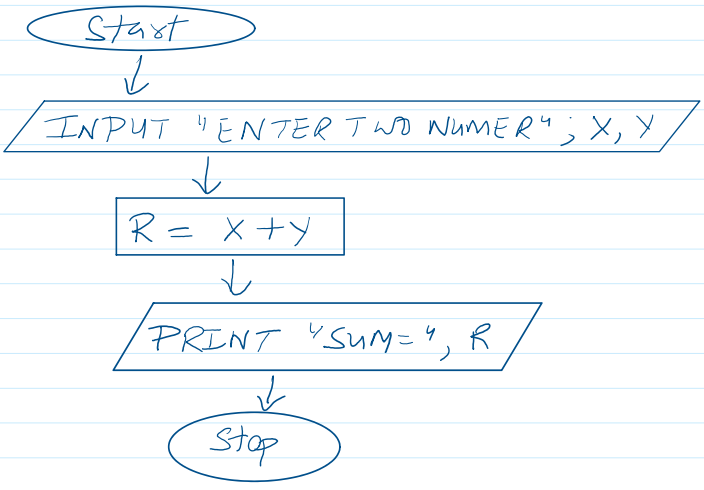
algorithms.

Flowchart- Visual Representation of Algorithm

A flowchart is a diagram made up of boxes, diamonds and other shapes connected by arrows. Each shape represents a step of the Solution process and the arrow represents the order or link among the steps.



① Draw a flow chart, to add two numbers entered by the user.



Home Work:

1. Two friends decide who gets the last slice of a cake by flipping a coin five times. The first person to win three flips wins the cake. An input of one means player one wins the flip. And two. means player two wins the flip design. Design of flow chart to determine who takes the cake.
2. Draw the flowchart to print all the multiples of five between 10 and 50 including both 10 and 50.
3. Suppose you are collecting money for something you need. rupees 200 in all. You ask your parents, uncles and aunts, as well as grandparents. Different people may give either rupees ₹10, 20, or even rupees 50. You will collect till the total becomes 200. Draw the flow chart.
4. Draw a flowchart to print the bill depending upon the price and quantity of an item. Also, print, bill GST , which is the bill after adding 5% of tax in the total bill.
5. Draw the flowchart that will perform the following task:
   a. Read the marks of three subjects computer science mathematics and physics out of 100.
   b. Calculate the aggregate marks.
   c. Calculate the percentage of marks.

6. Write an algorithm to find the greatest amount 2 different numbers entered by the user.